Aaditya Tamrakar

Software Engineer · Full Stack · Data Platforms · Algorithmic Trading

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Education

Master of Science, Computer Science CGPA 3.7/4 · 2021 – 2023

University of Massachusetts, Boston

- Coursework: Artificial Intelligence, Compilers, Computer Networks, Object Oriented Design and Programming, Software Development, User Interface Design, Theory of Computation.
- Graduate Teaching Fellow, CS240 Programming in C, for two semesters to classes of 51 and 67 students.

Work Experience

Viasat Inc. *Software Engineer II* Remote, Austin, US · May 2023 – Dec 2025

- Improved in-flight connectivity with the core platform team; developed a Golang microservice for a server deployed in airplanes for monitoring.
- Added new API endpoints, changes to the Snowflake database, and updated Airflow DAGs.
- Modified the CloudFormation stack to create new infrastructure for data monitoring.

Viasat Inc. *Software Engineer Intern* Remote, Boston, US · May – Aug 2022

- Consolidated functions into a common shared library used across teams; achieved 100% unit test coverage.
- Modified the cloud stack to create new infrastructure for data monitoring; joined full time after graduation.

Algo Maker LLP *Full Stack Developer* Mumbai, IN · Dec 2020 – Aug 2021

- Created and implemented 80+ efficient algorithms to execute trade orders, from time/price based to advanced market making using WebSockets, so it could scale to handle millions in funds.
- Designed a dashboard using Angular and Express.js to visualise data stored in MongoDB, which reduced the time to evaluate risk exposure and execute new orders by 50%.
- Set up the complete platform in AWS, closest to the broker's servers, resulting in 20% faster order execution.

Accio.ai *Software Engineer Intern* Mumbai, IN · Apr 2018 – Sep 2018

- Developed and implemented a data pipeline to process 2 million events per day.
- Designed a REST API to query data for ad payment and reporting.
- Created a searchable schema in MongoDB on the raw analytics data (clicks, impressions, views).
- Developed an automation script using the AWS SDK which cut the time to upload each design from minutes to seconds.

Freelance *Full Stack Developer* 2016 – 2019

- Shipped PHP/Laravel backends, Node.js APIs and Ionic/AngularJS apps for small businesses: food delivery, fleet management, dealership reporting, accounting.

Projects

Option Chain Analysis · Node.js, REST API 100+ stars on GitHub · Nov 2019

Fetches the live NSE Nifty option chain and classifies every strike by price and open interest change; used by 50+ developers in their own projects.

UMB Trading App · HTML5, JavaScript, Figma, User Interface Design · project site Dec 2021

CS615 project: a simple interface where a beginner can trade options easily.

ML Captcha Solver · scikit-learn, PHP Jan 2018

Image pipeline and classifier that reads Amazon-style captchas, 99.7% reported accuracy on that style.

Open Source

Forked, adapted and maintained broker SDKs for my own tooling: Kite Connect (Go), XTS Interactive v2 and 5paisa (Node.js), Hummingbot remote client (Python), option pricing formulas. Open sourced the option chain analyser, a futures spread chart and a Dream11 team builder.

Skills

Languages: Golang, Python, JavaScript, TypeScript, PHP, SQL, C, C++, Java

Technologies: Linux, Git, AWS, Docker, Kubernetes, Redis, MongoDB, Snowflake, Apache Airflow, Kafka, Spark, Playwright, Node.js, Angular, Laravel

Certifications: Crash Course on Python by Google (Coursera); HackerRank certificates in Problem Solving, JavaScript, Python, Node.js, CSS, REST API, SQL